PROBLEM STATEMENT

Agricultural yield is influenced by several environmental and agricultural factors such as rainfall, temperature, humidity, soil conditions and fertilizer usage. Traditional yield estimation methods may not effectively capture complex relationships between these variables.

This project aims to develop a machine learning model that predicts crop yield using environmental and agricultural features, with Gradient Boosted Trees optimized using Bayesian Hyperparameter Search.

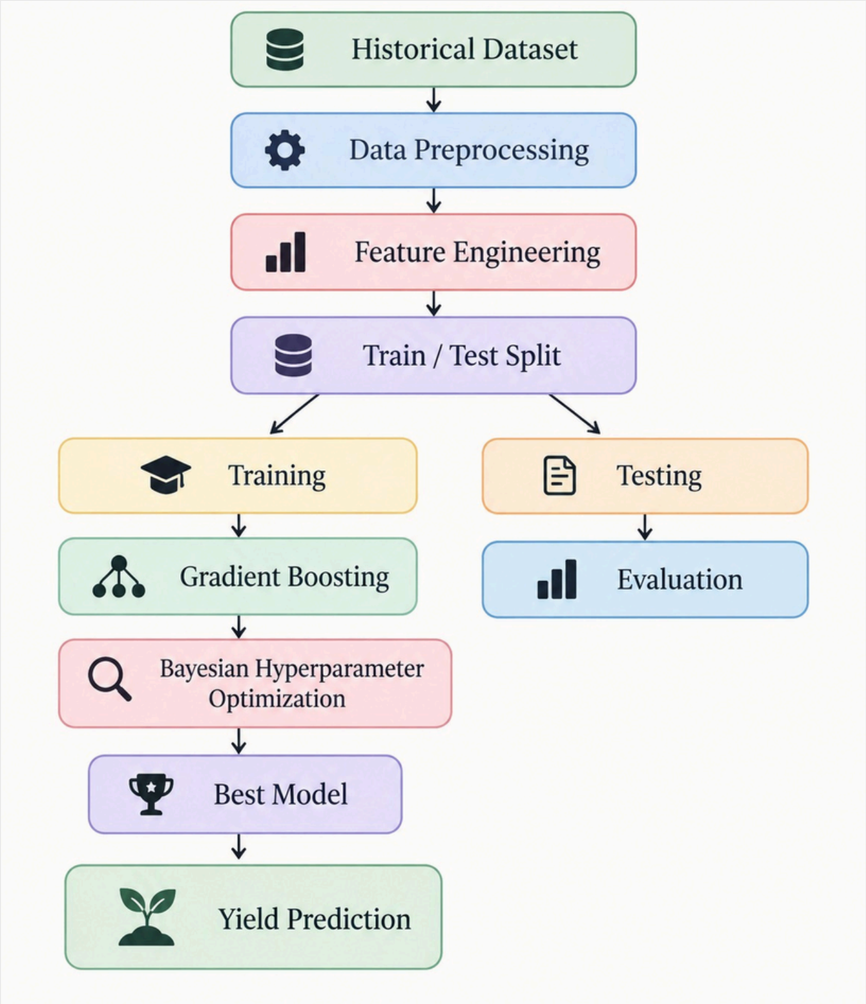
OBJECTIVES

- Collect historical agricultural and environmental data.
- Clean and preprocess the dataset.
- Identify important factors affecting crop yield.
- Develop a Gradient Boosted Tree regression model.
- Optimize model hyperparameters using Bayesian optimization.
- Evaluate the model using regression metrics.
- Predict crop yield for new input conditions.

DATASET AND FEATURES

Feature	Description
Rainfall	Total rainfall received
Temperature	Average temperature
Humidity	Average humidity
Soil Moisture	Moisture level of soil
Fertilizer	Fertilizer quantity
Crop Type	Type of crop
Region	Geographical region
Yield	Target variable

SYSTEM ARCHITECTURE



TOPICS USED

CO	Topics Used
CO1	ML Lifecycle, Supervised Learning
CO2	Preprocessing, Feature Engineering
CO3	Gradient Boosting, Feature Importance
CO4	K-Means Clustering, Pattern Discovery
CO5	RMSE, MAE, R^2, Bayesian Optimization
CO6	Model Training, Interpretation, ML Workflow

CONCLUSION & FUTURE SCOPE

The proposed system uses rainfall and other agricultural and environmental features to forecast crop yield using Gradient Boosted Trees. Bayesian Hyperparameter Optimization is used to systematically identify suitable model parameters. The model is evaluated using RMSE, MAE and R^2, while feature importance helps identify influential factors affecting yield.

Real-time weather data

Satellite imagery

Larger regional datasets

IoT-based soil sensors

Weather API integration

Model monitoring and automatic retraining

THANK YOU